Linear Algebra

Lecture 1: Systems of Linear Equations

Delivered in three videos | Based on Anton, *Elementary Linear Algebra*, §1.1

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Our six signature systems (used in every part of this lecture):

	System	Answer	Picture
U2	$x + y = 5,\ 2x - y = 4$	$(3, 2)$ unique	lines cross
N2	$x + y = 4,\ 2x + 2y = 6$	no solution	parallel lines
I2	$x + 2y = 4,\ 2x + 4y = 8$	infinitely many	same line
U3	$x + y + z = 6,\ x - y + z = 2,\ 2x + y - z = 1$	$(1, 2, 3)$ unique	planes meet at a point
I3	$x + y + z = 6,\ x - y + z = 2,\ 2x + 2z = 8$	infinitely many	planes share a line
N3	$x + y + z = 6,\ x - y + z = 2,\ 2x + 2z = 5$	no solution	“tent”

1 Video 1 — Systems and Their Solutions

1.1 A warm-up: the canteen problem

Two samosas and one chai cost Rs 70. One samosa and one chai cost Rs 50. Let s be the price of a samosa and c the price of a chai:

$$2s + c = 70, \qquad s + c = 50.$$

Subtracting the second equation from the first removes c: $s = 20$, and then $c = 30$. **That is already linear algebra.**

1.2 What is linear algebra?

In my own words

Linear algebra is the mathematics of straight lines and flat surfaces, handled many at a time, using tables of numbers called *matrices*.

It is the language behind computer graphics, search engines, data science and machine learning, because computers are very good at working with tables of numbers.

1.3 Linear equations

In my own words

An equation is *linear* if every variable appears only to the power 1: no squares, no products like xy, no roots, no $\sin x$, $\log x$ or e^x.

Definition: Linear equation

A linear equation in the n variables $x_1, \ldots, x_n$ is an equation that can be written as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, \ldots, a_n$ and b are constants and the a's are not all zero. If $b = 0$ the equation is called *homogeneous*.

$x + 3y = 7$	linear	$x^2 + y = 4$	not linear (square)
$\frac{1}{2}x - y + 3z = -1$	linear	$3x + 2y - xy = 5$	not linear (product)
$x_1 + x_2 + \cdots + x_n = 1$	linear	$\sin x + y = 0$	not linear (sine)

Common mistake

Constants may look scary but are fine: $\sqrt{2}\,x_1 + \pi x_2 = 7^{1/3}$ *is* linear. Only the *variables* must be "plain".

1.4 Systems and solutions

In my own words

A *system* is a group of linear equations that must all be true *at the same time*. A *solution* is a set of numbers that makes *every* equation true.

Definition: Solution

A solution of a linear system in $x_1, \ldots, x_n$ is a list of numbers $(s_1, \ldots, s_n)$ such that substituting $x_1 = s_1, \ldots, x_n = s_n$ makes each equation a true statement. We write it as an ordered pair (2 unknowns), triple (3 unknowns) or n-tuple.

Check for the canteen: $2(20) + 30 = 70$ ✓ and $20 + 30 = 50$ ✓, so $(s, c) = (20, 30)$.

1.5 Consistency and the three types of answers

Definition: Consistent / inconsistent

A system is **consistent** if it has *at least one* solution, and **inconsistent** if it has *no* solution.

Key fact

Every system of linear equations has **exactly one**, **infinitely many**, or **no** solutions. There are no other possibilities (never exactly two!).

Unique and infinite are *consistent*; only no solution is *inconsistent*.

Common mistake

"Infinitely many solutions" is **consistent**. Students often wrongly think consistent means "exactly one".

1.6 Three methods for two unknowns (system U2)

$$x + y = 5 \quad (1), \qquad 2x - y = 4 \quad (2)$$

Substitution. From (1), $y = 5 - x$. Put into (2): $2x - (5 - x) = 4 \Rightarrow 3x = 9 \Rightarrow x = 3$, so $y = 2$.

Elimination. Add (1) and (2): $3x = 9 \Rightarrow x = 3$; then from (1) $y = 2$.

Comparison. Make y the subject in both: $y = 5 - x$ and $y = 2x - 4$. Set them equal: $5 - x = 2x - 4 \Rightarrow x = 3$, $y = 2$.

All three give the **unique** solution $(3, 2)$: the system is consistent.

1.7 When elimination produces a contradiction (N2)

$$x + y = 4, \qquad 2x + 2y = 6.$$

Do $(2) - 2 \times (1)$: $0 = -2$. This is impossible, so there is **no solution**; the system is **inconsistent**. (Substitution gives the same: $y = 4 - x$ into (2) gives $8 = 6$.)

1.8 When elimination produces $0 = 0$ (I2)

$$x + 2y = 4, \qquad 2x + 4y = 8.$$

Do $(2) - 2 \times (1)$: $0 = 0$, always true. Equation (2) is just (1) doubled, so only one condition remains: $x + 2y = 4$. Let $y = t$ (a *parameter*, any real number). Then

$$x = 4 - 2t, \qquad y = t, \qquad t \in \mathbb{R}.$$

For example, $t = 0$ gives $(4, 0)$, $t = 1$ gives $(2, 1)$, $t = 2$ gives $(0, 2)$. **Infinitely many** solutions; the system is consistent.

1.9 Three unknowns

The strategy is the same: *eliminate one variable at a time* until one equation in one unknown remains.

U3: $x + y + z = 6$ **(1)**, $x - y + z = 2$ **(2)**, $2x + y - z = 1$ **(3)**

$(1) - (2)$: $2y = 4 \Rightarrow y = 2$.
Put $y = 2$ in (1) and (3): $x + z = 4$ and $2x - z = -1$.
Add these: $3x = 3 \Rightarrow x = 1$, then $z = 3$.
Unique solution $(1, 2, 3)$. Check in (3): $2 + 2 - 3 = 1$ ✓

I3: $x + y + z = 6$, $x - y + z = 2$, $2x + 2z = 8$

$(1) - (2)$: $y = 2$, so $x + z = 4$. The third equation is $(1) + (2)$, so it gives $2x + 2z = 8$, i.e. again $x + z = 4$: nothing new.
Let $z = t$: $x = 4 - t$, $y = 2$, $z = t$. **Infinitely many** solutions.

N3: $x + y + z = 6$, $x - y + z = 2$, $2x + 2z = 5$

$(1) + (2)$: $2x + 2z = 8$. But (3) says $2x + 2z = 5$. Subtracting: $0 = 3$, a contradiction. **No solution**; inconsistent.

2 Video 2 — What Solutions Look Like

2.1 Two variables: every equation is a line

In my own words

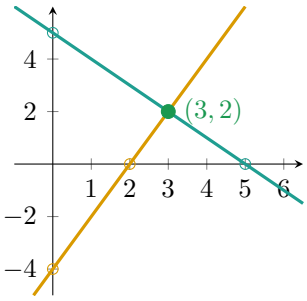
With two variables, the graph of a linear equation is a straight line. A solution of the system is a point that lies on *all* the lines, i.e. a point where they meet.

Sketching a line by hand (intercept method).

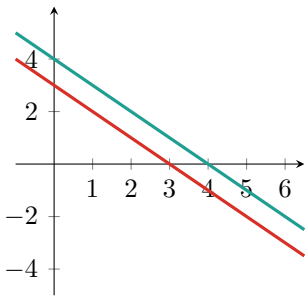
1. Put $x = 0$ and solve for y: this is where the line cuts the y-axis.
2. Put $y = 0$ and solve for x: this is where the line cuts the x-axis.
3. Join the two points with a ruler. (If the line passes through the origin, use one extra point instead.)

For U2: $x + y = 5$ passes through $(0, 5)$ and $(5, 0)$; $2x - y = 4$ passes through $(0, -4)$ and $(2, 0)$.

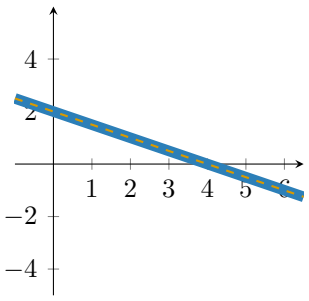
U2: cross → unique



N2: parallel → none



I2: same line → infinite



Algebra gives	Lines are	Number of solutions	Consistent?
a value for x and y	crossing	one	yes
$0 = 0$	the same line (coincident)	infinitely many	yes
$0 =$ nonzero	parallel and distinct	none	no

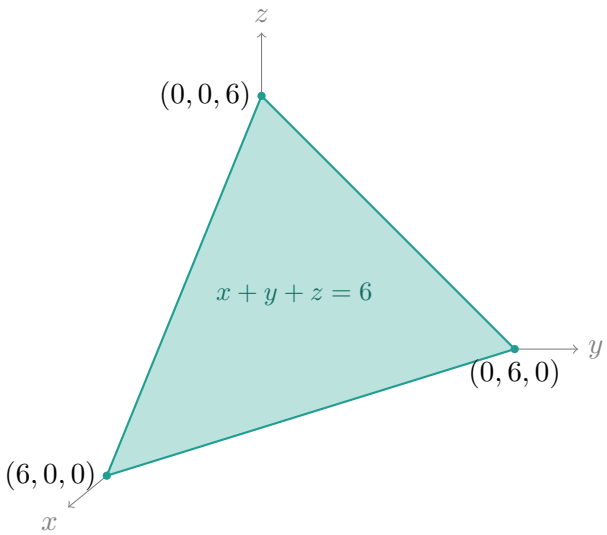
Quick test without solving: same slope + different intercept ⇒ parallel; one equation a multiple of the other ⇒ same line.

2.2 Three variables: every equation is a plane

In my own words

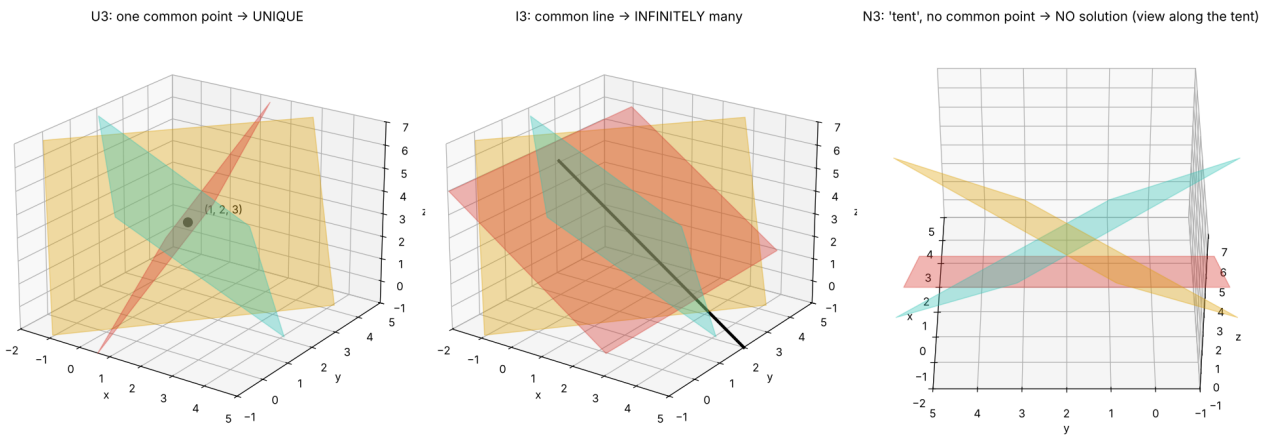
With three variables, the graph of a linear equation is a flat plane in space. A solution is a point that lies on all three planes at once.

Sketching a plane by hand. Find the three intercepts and join them into a triangle; that triangle is the visible piece of the plane in the first octant. For $x+y+z=6$ the intercepts are $(6,0,0)$, $(0,6,0)$, $(0,0,6)$:



For three planes the possibilities are:

- **one point** (U3): like the corner of a room, where two walls meet the floor;
- **a common line** (I3): like pages of a book meeting at the spine; or all planes the same;
- **no common point** (N3): e.g. a "tent" where each pair meets but all three never meet, or parallel planes.



Systems U3, I3 and N3 (from the Python notebook). In I3 the black line is $(4 - t,\ 2,\ t)$.

3 Video 3 — Matrices and Elementary Row Operations

3.1 The matrix form $AX = b$

In my own words

A computer does not need the letters x, y, z, the $+$ signs or the $=$ sign. It only needs the numbers, arranged in the right places.

For U3:

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}}_{A\ \text{(coefficients)}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{X\ \text{(unknowns)}} = \underbrace{\begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}}_{b\ \text{(constants)}} \quad \Longrightarrow \quad [A \mid b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right].$$

Definition: Augmented matrix

For a system with coefficient matrix A and right-hand side b, the *augmented matrix* is $[A \mid b]$: the coefficients with the constants attached as an extra column. Each **row** is one equation; each **column** (left of the bar) belongs to one unknown.

3.2 Elementary row operations (ERO)

Definition: Elementary row operations

1. **Interchange** two rows: $R_i \leftrightarrow R_j$.
2. **Multiply** a row by a *nonzero* constant: $R_i \to kR_i, \quad k \neq 0$.
3. **Add** a multiple of one row to another: $R_i \to R_i + kR_j$.

Why don't they change the solution? Each one is exactly what we already do to equations (reorder them, scale one, add a multiple of one to another), and each can be undone by another ERO. So the new system has exactly the same solutions as the old one.

Common mistake

Multiplying a row by 0 is *not* allowed: it wipes out an equation and can create new "solutions".

3.3 Solving U3 with row operations

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & -1 & -3 & -11 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -3 & -9 \end{array}\right] \xrightarrow{R_3 \to -\frac{1}{3}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Read off: $x = 1,\ y = 2,\ z = 3$ — the same point where the three planes meet.

3.4 Book Example 6 (with fractions)

$x + y + 2z = 9, \quad 2x + 4y - 3z = 1, \quad 3x + 6y - 5z = 0.$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right] \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \xrightarrow{\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array}\right] \xrightarrow{R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array}\right]$$

$$\xrightarrow{-2R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow[R_2 + \frac{7}{2}R_3]{R_1 - \frac{11}{2}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Solution: $(1, 2, 3)$.

3.5 Reading the last row

Last row looks like	Means	Type
$[0\ \ 0\ \ 0 \mid c]$ with $c \neq 0$	$0 = c$, impossible	no solution (inconsistent)
$[0\ \ 0\ \ 0 \mid 0]$ (and no contradiction)	an equation disappeared	infinitely many
full "staircase" of leading 1's	every unknown pinned down	unique

3.6 Worked exercises (Exercise Set 1.1)

Ex. 7(a) — write the augmented matrix: $-2x_1 = 6,\ 3x_1 = 8,\ 9x_1 = -3$

One unknown, three equations: $\left[\begin{array}{c|c} -2 & 6 \\ 3 & 8 \\ 9 & -3 \end{array}\right]$.

Ex. 9(a),(b) — is the triple a solution of $2x_1 - 4x_2 - x_3 = 1,\ x_1 - 3x_2 + x_3 = 1,\ 3x_1 - 5x_2 - 3x_3 = 1$**?**

(a) $(3, 1, 1)$: $6 - 4 - 1 = 1$ ✓, $3 - 3 + 1 = 1$ ✓, $9 - 5 - 3 = 1$ ✓. **Yes**, it is a solution.
(b) $(3, -1, 1)$: $6 + 4 - 1 = 9 \neq 1$. **No** (one failed equation is enough).

Ex. 11(a) — $3x - 2y = 4,\ 6x - 4y = 9$

$\left[\begin{array}{cc|c} 3 & -2 & 4 \\ 6 & -4 & 9 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 3 & -2 & 4 \\ 0 & 0 & 1 \end{array}\right]$. Last row says $0 = 1$: **no solution**; the lines are parallel (zero points of intersection).

Ex. 11(b) — $2x - 4y = 1,\ 4x - 8y = 2$

$\left[\begin{array}{cc|c} 2 & -4 & 1 \\ 4 & -8 & 2 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 2 & -4 & 1 \\ 0 & 0 & 0 \end{array}\right]$. Only $2x - 4y = 1$ remains. Let $y = t$: $x = \frac{1}{2} + 2t,\ y = t$. **Infinitely many**; the lines coincide.

Ex. 11(c) — $x - 2y = 0,\ x - 4y = 8$

$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 1 & -4 & 8 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & -2 & 8 \end{array}\right] \Rightarrow\ y = -4,\ x = 2y = -8$. **Unique**: the lines meet at $(-8, -4)$.

Ex. 15(b) — $x_1 + 3x_2 - x_3 = -4,\ 3x_1 + 9x_2 - 3x_3 = -12,\ -x_1 - 3x_2 + x_3 = 4$

$R_2 - 3R_1$ and $R_3 + R_1$ turn rows 2 and 3 into $[0\ 0\ 0 \mid 0]$. Only $x_1 + 3x_2 - x_3 = -4$ remains (three identical planes). With two parameters $x_2 = s$, $x_3 = t$:

$$x_1 = -4 - 3s + t, \quad x_2 = s, \quad x_3 = t.$$

Ex. 19(a) — **for which k is** $\left[\begin{array}{cc|c} 1 & k & -4 \\ 4 & 8 & 2 \end{array}\right]$ **consistent?**

$R_2 - 4R_1$: $[0\ \ 8 - 4k \mid 18]$. If $k = 2$ this row reads $0 = 18$: inconsistent. If $k \neq 2$ we can solve for y and then x. **Consistent for all $k \neq 2$.**

True/False (b) and (c)

(b) "Multiplying a row by zero is an ERO." **False**: the constant must be nonzero.
(c) "$x - y = 3,\ 2x - 2y = k$ cannot have a unique solution." **True**: $R_2 - 2R_1$ gives $0 = k - 6$, so either no solution ($k \neq 6$) or infinitely many ($k = 6$), never exactly one.

4 Exercise Set 1.1 — Complete Solutions

Students: try each problem first, then check. All answers were verified with exact (fraction) arithmetic.

Exercise 1 — linear in x_1, x_2, x_3?

(a)	$x_1 + 5x_2 - \sqrt{2}\,x_3 = 1$	linear	$\sqrt{2}$ is only a constant
(b)	$x_1 + 3x_2 + x_1x_3 = 2$	not linear	product x_1x_3
(c)	$x_1 = -7x_2 + 3x_3$	linear	rearranges to $x_1 + 7x_2 - 3x_3 = 0$
(d)	$x_1^{-2} + x_2 + 8x_3 = 5$	not linear	power -2
(e)	$x_1^{3/5} - 2x_2 + x_3 = 4$	not linear	power $3/5$
(f)	$\pi x_1 - \sqrt{2}\,x_2 = 7^{1/3}$	linear	π, $\sqrt{2}$, $7^{1/3}$ are constants

Exercise 2 — linear in x and y?

(a) $2^{1/3}x + \sqrt{3}\,y = 1$	linear	constant coefficients
(b) $2x^{1/3} + 3\sqrt{y} = 1$	not linear	roots of variables
(c) $\cos(\frac{\pi}{7})x - 4y = \log 3$	linear	$\cos(\pi/7)$ is a number
(d) $\frac{\pi}{7}\cos x - 4y = 0$	not linear	$\cos x$ of a variable
(e) $xy = 1$	not linear	product
(f) $y + 7 = x$	linear	$x - y = 7$

Exercise 3 — general systems

(a) $a_{11}x_1 + a_{12}x_2 = b_1, \quad a_{21}x_1 + a_{22}x_2 = b_2$

(b) $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \quad a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \quad a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$

(c) $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = b_1, \quad a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 = b_2$

Exercise 4 — their augmented matrices

(a) $\left[\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{array}\right]$ (b) $\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$ (c) $\left[\begin{array}{cccc|c} a_{11} & a_{12} & a_{13} & a_{14} & b_1 \\ a_{21} & a_{22} & a_{23} & a_{24} & b_2 \end{array}\right]$

Exercises 5–6 — system from an augmented matrix

Rule: the last column is the right-hand side; every other column belongs to one unknown.

5(a) $2x_1 = 0, \quad 3x_1 - 4x_2 = 0, \quad x_2 = 1$

5(b) $3x_1 - 2x_3 = 5, \quad 7x_1 + x_2 + 4x_3 = -3, \quad -2x_2 + x_3 = 7$

6(a) $3x_2 - x_3 - x_4 = -1, \quad 5x_1 + 2x_2 - 3x_4 = -6$

6(b) $3x_1 + x_3 - 4x_4 = 3, \quad -4x_1 + 4x_3 + x_4 = -3, \quad -x_1 + 3x_2 - 2x_4 = -9, \quad -x_4 = -2$

Exercises 7–8 — augmented matrix from a system

Rule: a missing variable gets a 0 in its column.

7(a) $\left[\begin{array}{c|c} -2 & 6 \\ 3 & 8 \\ 9 & -3 \end{array}\right]$ 7(b) $\left[\begin{array}{ccc|c} 6 & -1 & 3 & 4 \\ 0 & 5 & -1 & 1 \end{array}\right]$ 7(c) $\left[\begin{array}{ccccc|c} 0 & 2 & 0 & -3 & 1 & 0 \\ -3 & -1 & 1 & 0 & 0 & -1 \\ 6 & 2 & -1 & 2 & -3 & 6 \end{array}\right]$

8(a) $\left[\begin{array}{cc|c} 3 & -2 & -1 \\ 4 & 5 & 3 \\ 7 & 3 & 2 \end{array}\right]$ 8(b) $\left[\begin{array}{ccc|c} 2 & 0 & 2 & 1 \\ 3 & -1 & 4 & 7 \\ 6 & 1 & -1 & 0 \end{array}\right]$ 8(c) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$

Exercise 9 — $2x_1 - 4x_2 - x_3 = 1, \;\; x_1 - 3x_2 + x_3 = 1, \;\; 3x_1 - 5x_2 - 3x_3 = 1$

(a) $(3, 1, 1)$	$1,\ 1,\ 1$	solution
(b) $(3, -1, 1)$	first eq. gives $6 + 4 - 1 = 9 \neq 1$	not a solution
(c) $(13, 5, 2)$	first eq. gives $26 - 20 - 2 = 4 \neq 1$	not a solution
(d) $(\frac{13}{2}, \frac{5}{2}, 2)$	$13 - 10 - 2 = 1,\ \frac{13}{2} - \frac{15}{2} + 2 = 1,\ \frac{39}{2} - \frac{25}{2} - 6 = 1$	solution
(e) $(17, 7, 5)$	$34 - 28 - 5 = 1,\ 17 - 21 + 5 = 1,\ 51 - 35 - 15 = 1$	solution

Note: three different solutions means this system has *infinitely many* solutions (it can never have exactly three).

Exercise 10 — $x + 2y - 2z = 3,\ \ 3x - y + z = 1,\ \ -x + 5y - 5z = 5$

(a) $(\frac{5}{7}, \frac{8}{7}, 1)$	first eq. gives $\frac{21}{7} - 2 = 1 \neq 3$	not a solution
(b) $(\frac{5}{7}, \frac{8}{7}, 0)$	3, 1, 5	solution
(c) $(5, 8, 1)$	first eq. gives $19 \neq 3$	not a solution
(d) $(\frac{5}{7}, \frac{10}{7}, \frac{2}{7})$	3, 1, 5	solution
(e) $(\frac{5}{7}, \frac{22}{7}, 2)$	3, 1, 5	solution

Exercise 11

Solved step by step in §3.6: (a) no solution (parallel lines); (b) infinitely many, $x = \frac{1}{2} + 2t,\ y = t$ (same line); (c) unique, $(-8, -4)$.

Exercise 12 — $2x - 3y = a,\ \ 4x - 6y = b$

$R_2 - 2R_1$: $\quad [\,0\ \ 0 \mid b - 2a\,]$.

- $b \neq 2a$: last row says $0 = b - 2a \neq 0 \ \Rightarrow$ no solution (parallel lines).
- $b = 2a$: last row is all zeros $\Rightarrow$ infinitely many (same line).
- One solution: never, for any a, b (the two lines always have the same slope).

Exercises 13–14 — parametric description of one equation

Method: solve for the first variable; every other variable becomes a free parameter.

13(a) $x = \frac{3}{7} + \frac{5}{7}t,\ \ y = t$
13(b) $x_1 = \frac{7}{3} + \frac{5}{3}r - \frac{4}{3}s,\ \ x_2 = r,\ \ x_3 = s$
13(c) $x_1 = -\frac{1}{8} + \frac{1}{4}r - \frac{5}{8}s + \frac{3}{4}t,\ \ x_2 = r,\ x_3 = s,\ x_4 = t$
13(d) $v = \frac{8}{3}r - \frac{2}{3}s + \frac{1}{3}t - \frac{4}{3}u,\ \ w = r,\ x = s,\ y = t,\ z = u$

14(a) $x = 2 - 10t,\ \ y = t$
14(b) $x_1 = 3 - 3r + 12s,\ \ x_2 = r,\ \ x_3 = s$
14(c) $x_1 = 5 - \frac{1}{2}r - \frac{3}{4}s - \frac{1}{4}t,\ \ x_2 = r,\ x_3 = s,\ x_4 = t$
14(d) $v = -r - s + 5t - 7u,\ \ w = r,\ x = s,\ y = t,\ z = u$

Exercises 15–16 — systems with infinitely many solutions

In each system every equation is a multiple of one equation, so only one condition remains.

15(a) $(2) = 3 \times (1)$: $\ 2x - 3y = 1 \ \Rightarrow\ x = \frac{1}{2} + \frac{3}{2}t,\ \ y = t$
15(b) $(2) = 3 \times (1),\ (3) = -1 \times (1)$: $\ x_1 = -4 - 3r + s,\ \ x_2 = r,\ \ x_3 = s$
16(a) $(1) = 2 \times (2)$: $\ 3x_1 + x_2 = -4 \ \Rightarrow\ x_1 = t,\ \ x_2 = -4 - 3t$
16(b) $(2) = 3 \times (1),\ (3) = -2 \times (1)$: $\ 2x - y + 2z = -4 \ \Rightarrow\ x = r,\ \ y = 4 + 2r + 2s,\ \ z = s$

Exercises 17–18 — one ERO that puts a 1 in the top-left corner (no fractions in row 1)

17(a)	$R_1 \to R_1 + 2R_2$	new $R_1 = [\,1\ \ -7\ \ 8\ \ 8\,]$
17(b)	$R_1 \leftrightarrow R_3$	new $R_1 = [\,1\ \ 4\ \ -3\ \ 3\,]$
18(a)	$R_1 \to \frac{1}{2}R_1$	new $R_1 = [\,1\ \ 2\ \ -3\ \ 4\,]$ (all entries were even)
18(b)	$R_1 \to R_1 - 2R_2$	new $R_1 = [\,1\ \ -2\ \ -18\ \ 0\,]$ (or $R_1 + R_3 = [\,1\ \ -1\ \ -3\ 6\,]$)

Exercises 19–20 — values of k for a consistent system

19(a)	$R_2 - 4R_1$: $[\,0 \;\; 8-4k \mid 18\,]$	$k=2$ gives $0=18$	consistent for $k \neq 2$
19(b)	$R_2 - 4R_1$: $[\,0 \;\; 8-4k \mid 0\,]$	never a contradiction	consistent for all k
20(a)	$R_2 + 2R_1$: $[\,0 \;\; 0 \mid 5+2k\,]$	needs $5+2k=0$	consistent only for $k=-\frac{5}{2}$
20(b)	$k \neq -4$: unique; $k=-4$: $R_2+R_1=[\,0\;0 \mid 0\,]$	never a contradiction	consistent for all k

Exercise 21 — parabola through three points

If (x_i, y_i) lies on $y = ax^2 + bx + c$, then $ax_i^2 + bx_i + c = y_i$ for $i = 1, 2, 3$. These are three linear equations in the unknowns a, b, c, with coefficients x_i^2, x_i, 1 and right-hand side y_i, which is exactly the given augmented matrix.

Exercise 22 — why EROs do not change the solution set

Each operation can be undone by another operation of the same kind: a swap undoes itself; $R_i \to kR_i$ is undone by $R_i \to \frac{1}{k}R_i$ (this is why $k \neq 0$); $R_i \to R_i + kR_j$ is undone by $R_i \to R_i - kR_j$. Any solution of the old system satisfies the new one (we only combined true equations), and because the step can be reversed, any solution of the new system satisfies the old one. So the two solution sets are equal.

Exercise 23

Take $x_2 = 0$: then $x_1 = c$ solves the first equation, so it must solve the second: $c = d$. Take $x_2 = 1$: then $x_1 = c - k$ solves the first, so $c - k + l = d = c$, giving $k = l$. Hence the equations are identical.

Exercise 24 — three lines in the plane

(a) No solution: the three lines have no common point (e.g. at least two are parallel and distinct, or they meet in pairs forming a triangle).
(b) Exactly one: all three lines pass through one common point (at least two of them are not parallel).
(c) Infinitely many: all three lines are the same line.

Exercise 25 — diet problem

Fat: $2x + 3y + z = 7$, protein: $2x + y + 3z = 9$, carbohydrates: $4x + 2y + 5z = 16$.

Exercise 26 — parabola through $(1,1), (2,4), (-1,1)$

$a + b + c = 1$, $4a + 2b + c = 4$, $a - b + c = 1$. We expect **exactly one** solution, because through three points with different x-values there passes exactly one parabola (or line). (Solving gives $a = 1$, $b = 0$, $c = 0$, i.e. $y = x^2$.)

Exercise 27

$x + y + z = 12$, $2x + y + 2z = 5$, $z = x + 1$ (i.e. $-x + z = 1$).

True / False

(a) **True** $x_1 = \cdots = x_n = 0$ always satisfies homogeneous equations.
(b) **False** the constant must be nonzero.
(c) **True** $R_2 - 2R_1$ gives $0 = k - 6$: none or infinitely many, never one.
(d) **True** solve for one variable with nonzero coefficient; the others are free.
(e) **False** e.g. $x = 1,\ 2x = 2,\ 3x = 3$ is consistent.
(f) **False** scaling equations does not change the solutions at all; they are not multiplied by c.
(g) **True** add -1 times one row to another.
(h) **False** the second row says $0 = -1$, so the system is inconsistent.

Working with Technology

T1. In Python: `np.linalg.solve(A, b)` returns $(\frac{7}{3}, \frac{4}{3})$ for Example 2, but raises `LinAlgError: Singular matrix` for Examples 3 and 4, because those systems do not have exactly one solution. The rank test in the notebook (`classify`) tells them apart: Example 3 is inconsistent, Example 4 has infinitely many solutions.

T2. Reading the points as $(-1, 14)$, $(0, 8)$, $(1, 7)$: $\quad c = 8, \quad a - b = 6, \quad a + b = -1$, so $a = \frac{5}{2},\ b = -\frac{7}{2},\ c = 8$.